Dear Editor,

We are pleased to submit our Article, “Warming and survey effort contribute comparably to new bird distribution records when dated to discovery”, for consideration in *Nature Communications*.

New provincial and national distribution records are among the most widely used evidence that species are shifting their ranges under climate change. Yet every record needs a species to be present, an observer to look and a paper to be written, and in China, as in much of the world, warming and citizen-science effort have grown together: annual birding visits in the main checklist databases rose about 175-fold between 2002–2004 and 2022–2024. Analyses that include only one of these processes credit it with both. Most also date records to their year of publication, a choice that, we show, changes the conclusions.

Using the China Bird New Record dataset (CBNR; 1,029 provincial records, 2000–2025, openly available on Zenodo), we model the annual hazard of a first provincial record for every species in every province it could plausibly reach. Three findings stand out:

- **Dating is a modelling decision.** Publication-year dating inflates the survey-effort effect by 37% and manufactures a spurious climate-variability effect; dating to discovery removes both while leaving the warming effect intact.
- **Warming and observation contribute comparably.** Accumulated warming relative to each species’ historical range (hazard ratio 1.36 per standard deviation) and survey effort (1.40) have similar effects and interact negatively, and design-preserving permutation nulls quantify how much of the warming association is species-specific.
- **The observation process is large and saturating.** Province-by-year survey pulses are the largest variance component, and the effect of effort fell from a hazard ratio of 2.18 in 2002 to 1.07 in 2024, whereas the warming effect showed no such decline.

The same care changes how the direction of new records should be read: their apparent eastward bias is reproduced by geography alone, and once geography and survey effort are accounted for, a modest poleward excess remains.

We believe the work suits the journal’s broad, multidisciplinary readership: it connects biodiversity science, climate-change attribution, citizen science and the statistics of observational data, and it offers a transparent, reproducible template for separating ecological change from the history of observation, including explicit limits on how far such models can be projected under climate scenarios.

All data and code are openly available (CBNR: https://doi.org/10.5281/zenodo.20809949; analysis code: https://github.com/dingchenchen6/bird-newrecord-discovery-dated, to be archived with a DOI on acceptance).

A data descriptor for CBNR is under consideration at *Scientific Data*; it documents the dataset and contains none of the analyses reported here. [Authors: state any preprint.] This manuscript is not under consideration elsewhere, all authors have approved its submission, and we declare no competing interests.

[Optional: suggested and excluded reviewers, to be provided by the authors.]

Sincerely,

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